

4	(a) (i) (change in) potential energy = mgh = $0.056 \times 9.8 \times 16$ = 8.78 J (allow 8.8)	C1	A1	[2]
	(ii) (initial) kinetic energy = $\frac{1}{2}mv^2$ = $\frac{1}{2} \times 0.056 \times 18^2$ = 9.07 J (allow 9.1) total kinetic energy = $8.78 + 9.07 = 17.9$ J	C1	C1	A1
				[3]
	(b) kinetic energy = $\frac{1}{2}mv^2$ $17.9 = \frac{1}{2} \times 0.056 \times v^2$ and $v = 25(.3) \text{ m s}^{-1}$	B1		[1]
	(c) horizontal velocity = 18 m s^{-1}	B1		[1]
	(d) (i) correct shape of diagram (two sides of right-angled triangle with correct orientation)	B1		
	(ii) angle = $41^\circ \rightarrow 48^\circ$ (allow trig. solution based on diagram) (for angle $38^\circ \rightarrow 41^\circ$ or $48^\circ \rightarrow 51^\circ$, allow 1 mark)	A2		[3]

5	(a) (i) vibrations (in plane) normal to direction of energy propagation	B1		[1]
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